6.9 Polar Equations of Conics

Name _____ Period ____ Date _____

Pre-Calculus 6.9 · Student sheet

I can…

- find the eccentricity of a conic from its polar equation and identify it.
- find the vertices of a conic in polar form.
- write the polar equation of a conic.

Vocabulary · fill in the missing word

Eccentricity — e: the ratio of a point's distance to the focus and its distance to the _____

Directrix — the fixed line used with the focus to _____ a conic

Part 1 · Slide the eccentricity

Graph $r = e/(1 - e\cos\theta)$ with a slider for e.

e	0.5	1	2
Shape			

Part 2 · Notes and examples (fill in each note, then work the example)

Polar conics: $r = ep/(1 \pm e\cos\theta)$ or $r = ep/(1 \pm e\sin\theta)$: ellipse if $e < 1$, parabola if $e = 1$, hyperbola if e _____ 1.

Reading e: first rewrite so the denominator starts with _____.

Example 1

Identify the conic $r = \dfrac{15}{3 - 2\cos\theta}$.

Example 2

Identify $r = \dfrac{32}{3 + 5\sin\theta}$ and find its vertices.

Directrix: cos θ means a vertical directrix ($x = \pm p$); sin θ means a _____ directrix ($y = \pm p$).

Example 3

Write the polar equation of the parabola with focus at the pole and directrix $y = 3$.

Example 4

Find the eccentricity and vertices of $r = \frac{6}{2 + \cos\theta}$.

Part 3 · Partner practice

Partner 1

Identify $r = \frac{8}{4 + 4\sin\theta}$.

Partner 2

Write the polar equation of the ellipse with $e = \frac{1}{2}$, focus at the pole, and directrix $x = 4$.